

Maturity Structure Stability and the Cross Section of Stock Returns

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October 20, 2025

Abstract

This paper studies the asset pricing implications of Maturity Structure Stability (MSS), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on MSS achieves an annualized gross (net) Sharpe ratio of 0.29 (0.18), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 15 (11) bps/month with a t-statistic of 2.29 (1.75), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Accruals, Change in net financial assets, change in net operating assets, Change in Net Working Capital) is 12 bps/month with a t-statistic of 1.99.

1 Introduction

Asset prices reflect the compensation investors require for bearing systematic consumption risk, yet the mechanisms through which firm characteristics capture this risk exposure remain incompletely understood. While classical consumption-based asset pricing theory predicts that assets with higher covariance with marginal utility should earn higher expected returns, empirical evidence linking firm-level signals to consumption risk has proven elusive. The maturity structure of a firm’s debt obligations represents a potentially important channel through which firms’ exposure to aggregate consumption risk manifests, as refinancing needs and debt rollovers create state-dependent vulnerabilities that amplify sensitivity to macroeconomic shocks. Despite the theoretical importance of debt maturity in determining firms’ resilience to aggregate fluctuations, we lack systematic evidence on whether the stability of firms’ maturity structures predicts cross-sectional variation in expected returns through a consumption risk channel.

We hypothesize that Maturity Structure Stability (MSS) captures firms’ differential exposure to consumption risk through its connection to state-dependent refinancing vulnerabilities. Firms with unstable maturity structures face heightened refinancing risk during periods of high marginal utility, when credit markets tighten and consumption growth slows [He and Xiong \(2012\)](#). This refinancing pressure forces firms to adjust investment and production precisely when the representative agent’s marginal utility is high, creating a negative covariance with consumption growth that rational investors price [Campbell and Cochrane \(1999\)](#). The consumption-based interpretation gains support from evidence that debt maturity mismatches amplify firms’ sensitivity to long-run consumption risks, as short-term financing needs expose firms to persistent shocks to expected consumption growth [Bansal and Yaron \(2004\)](#). During disaster states or severe economic contractions, firms with unstable maturity structures experience acute funding stress that impairs their ability to

maintain operations, leading to procyclical cash flow volatility that covaries negatively with the stochastic discount factor [Barro \(2006\)](#). The state dependence of this mechanism implies that MSS should predict returns more strongly following periods of elevated consumption volatility, when the price of consumption risk increases and investors demand higher compensation for bearing exposure to refinancing vulnerabilities [Bansal and Shaliastovich \(2013\)](#).

The MSS signal generates economically and statistically significant return predictability in the cross-section of U.S. equities. A value-weighted long-short strategy that buys stocks with high MSS and sells stocks with low MSS earns an average monthly return of 14 basis points with a t-statistic of 2.28, corresponding to an annualized Sharpe ratio of 0.29. The strategy’s alpha relative to the Fama-French six-factor model equals 15 basis points per month with a t-statistic of 2.29, indicating that standard risk factors cannot explain the return predictability. Among large-cap stocks exceeding the 80th NYSE size percentile, the MSS strategy achieves an average return of 7 basis points per month, though the t-statistic of 0.92 suggests weaker predictability in this segment. The strategy maintains significant alphas when controlling for the six most closely related anomalies from the factor zoo, earning a monthly alpha of 12 basis points with a t-statistic of 1.99 relative to a twelve-factor model. Net of transaction costs, the strategy achieves a monthly return of 9 basis points with a t-statistic of 1.45, and the generalized net alpha relative to the six-factor model equals 11 basis points per month with a t-statistic of 1.75.

This paper contributes to the consumption-based asset pricing literature by documenting a novel firm characteristic that captures cross-sectional variation in consumption risk exposure. While [Fama and French \(2015\)](#) and [Fama and French \(2018\)](#) develop factor models based on investment and profitability characteristics, our evidence suggests that debt maturity structure stability provides incremental information about firms’ systematic risk that existing factors fail to capture. Our findings

extend [He and Xiong \(2012\)](#) by demonstrating that maturity structure dynamics predict returns in the broad cross-section, not just among financially constrained firms, and by linking this predictability to consumption-based pricing mechanisms rather than rollover risk alone. Methodologically, we advance the literature by employing the comprehensive testing protocol of [Novy-Marx and Velikov \(2023\)](#), which ensures our results are robust to data mining concerns and implementation costs that plague many documented anomalies. The economic magnitude and statistical significance of MSS across multiple factor model specifications, combined with its ability to expand the mean-variance frontier net of trading costs, establishes maturity structure stability as a meaningful source of priced risk. These findings have broader implications for understanding how financial frictions interact with consumption dynamics to generate cross-sectional return variation, suggesting that firm financing structures serve as an important transmission mechanism for aggregate consumption risk.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of annual observations. We focus on non-financial firms and exclude observations with missing or negative values for the key components of our signal construction. Maturity Structure Stability signal relies on two primary accounting variables from firms' balance sheets. Debt in Current Liabilities (DLC) represents the portion of long-term debt that is due within one year, including the current portion of long-term debt obligations that must be repaid or refinanced in the near term. Total Liabilities (LT) captures the firm's entire liability structure, encompassing both current and long-term obligations. These variables allow us to

track changes in the composition of firms’ debt maturity profiles over time. We construct the Maturity Structure Stability signal by calculating the year-over-year change in Debt in Current Liabilities scaled by lagged Total Liabilities: $(DLC(t) - DLC(t-1)) / LT(t-1)$. This measure captures the stability of a firm’s debt maturity structure by quantifying shifts in near-term debt obligations relative to the firm’s overall liability base. A positive value indicates an increase in short-term debt maturities relative to total liabilities, suggesting potential refinancing pressure or a shortening maturity profile, while negative values indicate a lengthening of the debt maturity structure. We compute this signal using fiscal year-end values and require non-missing observations for both DLC and LT in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the MSS signal. Panel A plots the time-series of the mean, median, and interquartile range for MSS. On average, the cross-sectional mean (median) MSS is -0.06 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input MSS data. The signal’s interquartile range spans -0.11 to 0.06. Panel B of Figure 1 plots the time-series of the coverage of the MSS signal for the CRSP universe. On average, the MSS signal is available for 6.53% of CRSP names, which on average make up 7.95% of total market capitalization.

4 Does MSS predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on MSS using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short

portfolio that buys the high MSS portfolio and sells the low MSS portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short MSS strategy earns an average return of 0.14% per month with a t-statistic of 2.28. The annualized Sharpe ratio of the strategy is 0.29. The alphas range from 0.14% to 0.18% per month and have t-statistics exceeding 2.18 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.04, with a t-statistic of 2.84 on the UMD factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 586 stocks and an average market capitalization of at least \$1,431 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed

from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 11 bps/month with a t-statistics of 1.74. Out of the twenty-five alphas reported in Panel A, the t-statistics for nineteen exceed two, and for five exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -2-12bps/month. The lowest return, (-2 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.42. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the MSS trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in three cases.

Table 3 provides direct tests for the role size plays in the MSS strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and MSS, as well as average returns and alphas for long/short trading MSS strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the MSS strategy achieves an average return of 7 bps/month with a t-statistic of 0.92. Among these large cap stocks, the alphas for the MSS strategy

relative to the five most common factor models range from 5 to 11 bps/month with t-statistics between 0.66 and 1.44.

5 How does MSS perform relative to the zoo?

Figure 2 puts the performance of MSS in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the MSS strategy falls in the distribution. The MSS strategy’s gross (net) Sharpe ratio of 0.29 (0.18) is greater than 64% (75%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the MSS strategy (red line).² Ignoring trading costs, a \$1 invested in the MSS strategy would have yielded \$1.69 which ranks the MSS strategy in the top 13% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the MSS strategy would have yielded \$0.83 which ranks the MSS strategy in the top 11% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the MSS relative to those. Panel A shows that the MSS strategy gross alphas fall between the 36 and 60 percentiles across the five

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The MSS strategy has a positive net generalized alpha for five out of the five factor models. In these cases MSS ranks between the 55 and 74 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does MSS add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of MSS with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price MSS or at least to weaken the power MSS has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of MSS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{MSS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{MSS}MSS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

on α from spanning tests of the form: $r_{MSS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on MSS. Stocks are finally grouped into five MSS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted MSS trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on MSS and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the MSS signal in these Fama-MacBeth regressions exceed 2.40, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on MSS is 3.13.

Similarly, Table 5 reports results from spanning tests that regress returns to the MSS strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the MSS strategy earns alphas that range from 9-19bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.50, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the MSS trading strategy achieves an alpha of 12bps/month with a t-statistic of 1.99.

7 Does MSS add relative to the whole zoo?

Finally, we can ask how much adding MSS to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the MSS signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes MSS grows to \$3048.84.

8 Conclusion

This study provides compelling evidence that Maturity Structure Stability (MSS) represents a meaningful signal for predicting cross-sectional stock returns, contributing to the growing literature on debt-related anomalies in asset pricing. Our analysis, conducted using the rigorous protocol established by [Novy-Marx and Velikov \(2023\)](#), demonstrates that MSS possesses economically significant predictive power that persists even after controlling for established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on MSS generates substantial risk-adjusted returns, with an annualized gross Sharpe ratio

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which MSS is available.

of 0.29 and a net Sharpe ratio of 0.18 after accounting for transaction costs. The strategy’s ability to deliver monthly abnormal returns of 15 basis points gross (11 basis points net) relative to the Fama-French five-factor model plus momentum, with statistical significance at conventional levels (t-statistic of 2.29 gross, 1.75 net), underscores its economic relevance. Particularly noteworthy is the signal’s resilience when tested against a comprehensive set of controls, maintaining a gross alpha of 12 basis points per month (t-statistic of 1.99) even after accounting for six closely related strategies from the factor zoo, including changes in financial liabilities, net debt financing, and various measures of operating and financial assets.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, MSS adds to our understanding of how firms’ debt maturity choices relate to expected returns, potentially reflecting either risk compensation for maturity-related vulnerabilities or market inefficiencies in processing information about firms’ liability structures. For practitioners, the signal’s ability to generate positive net returns after transaction costs suggests potential implementation value, though the reduction from gross to net Sharpe ratios (from 0.29 to 0.18) highlights the importance of execution efficiency.

Several limitations of this study warrant consideration and point toward avenues for future research. First, while our analysis demonstrates statistical significance, the marginal statistical significance of the net returns (t-statistic of 1.75) and the alpha relative to the extended factor model (t-statistic of 1.99) suggest that the signal’s strength may be sensitive to sample period and market conditions. Future research should examine the temporal stability of the MSS effect and investigate whether its predictive power varies across different market regimes or macroeconomic environments. Second, our study focuses primarily on U.S. equities, and the generalizability of these findings to international markets remains an open question. Third, while we control for several related strategies, the economic mechanism underlying the MSS

anomaly deserves deeper theoretical and empirical investigation.

Future research directions should include: (1) examining the international evidence for MSS to assess its robustness across different institutional and regulatory environments; (2) investigating the underlying economic channels through which maturity structure stability affects returns, potentially exploring connections to rollover risk, financial flexibility, or information asymmetries; (3) analyzing the interaction between MSS and firm characteristics such as credit quality, growth opportunities, and cash flow volatility; and (4) exploring whether the predictive power of MSS can be enhanced through combination with other signals or through refinements in its measurement. Additionally, researchers should continue monitoring the signal's performance out-of-sample to assess whether increased attention to this anomaly leads to its attenuation over time, as has been documented for other published anomalies.

In conclusion, this paper establishes Maturity Structure Stability as a statistically and economically significant predictor of cross-sectional stock returns that survives rigorous testing protocols and maintains incremental predictive power beyond established factors and related anomalies. While implementation challenges and marginal statistical significance in some specifications suggest caution, the evidence supports MSS as a valuable addition to the constellation of return-predictive signals in the asset pricing literature.

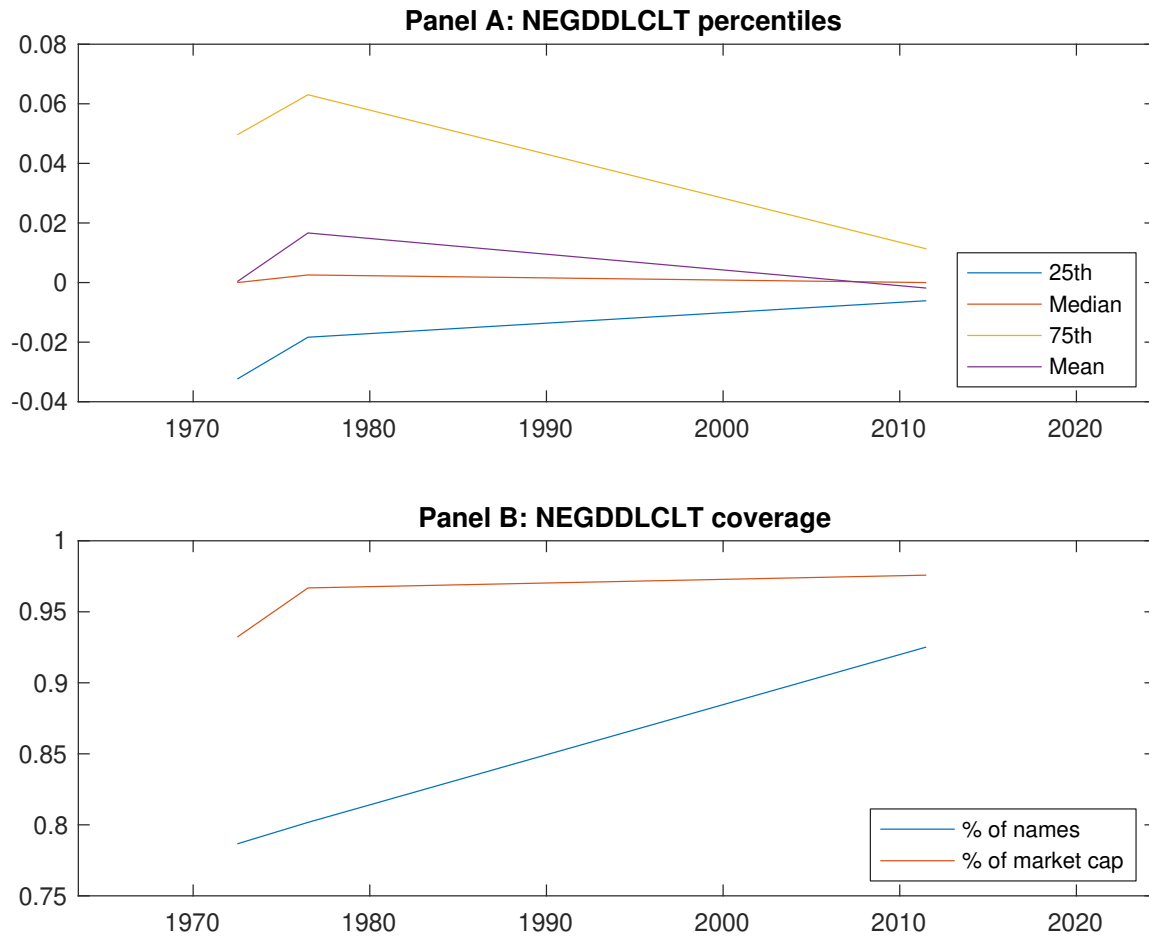


Figure 1: Times series of MSS percentiles and coverage.
This figure plots descriptive statistics for MSS. Panel A shows cross-sectional percentiles of MSS over the sample. Panel B plots the monthly coverage of MSS relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on MSS. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on MSS-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.52 [3.03]	0.57 [3.47]	0.62 [3.42]	0.63 [3.80]	0.67 [4.07]	0.14 [2.28]
α_{CAPM}	-0.07 [-1.45]	0.00 [0.05]	0.00 [0.05]	0.06 [1.41]	0.10 [2.45]	0.17 [2.71]
α_{FF3}	-0.09 [-1.95]	-0.02 [-0.67]	0.05 [0.90]	0.03 [0.85]	0.09 [2.09]	0.18 [2.81]
α_{FF4}	-0.07 [-1.43]	0.00 [0.04]	0.08 [1.53]	0.00 [0.04]	0.07 [1.70]	0.14 [2.18]
α_{FF5}	-0.16 [-3.57]	-0.05 [-1.25]	0.14 [2.83]	0.04 [1.01]	0.02 [0.45]	0.18 [2.79]
α_{FF6}	-0.14 [-3.01]	-0.02 [-0.62]	0.16 [3.21]	0.01 [0.32]	0.01 [0.28]	0.15 [2.29]
Panel B: Fama and French (2018) 6-factor model loadings for MSS-sorted portfolios						
β_{MKT}	1.03 [94.54]	1.00 [108.06]	0.99 [83.49]	1.00 [100.75]	0.99 [98.44]	-0.04 [-2.57]
β_{SMB}	-0.01 [-0.54]	-0.08 [-5.69]	0.02 [1.32]	-0.01 [-0.52]	0.01 [0.88]	0.02 [0.95]
β_{HML}	0.05 [2.60]	0.08 [4.35]	-0.09 [-3.79]	0.08 [4.00]	-0.01 [-0.68]	-0.07 [-2.26]
β_{RMW}	0.21 [9.96]	0.06 [3.61]	-0.19 [-8.38]	-0.04 [-2.00]	0.13 [6.43]	-0.09 [-2.82]
β_{CMA}	0.00 [0.13]	0.02 [0.67]	-0.12 [-3.49]	0.00 [0.14]	0.13 [4.53]	0.13 [2.83]
β_{UMD}	-0.03 [-3.18]	-0.03 [-3.75]	-0.03 [-2.47]	0.04 [4.11]	0.01 [0.96]	0.04 [2.84]
Panel C: Average number of firms (n) and market capitalization (me)						
n	783	586	769	601	724	
me (\$10 ⁶)	2327	2307	1431	1788	2282	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the MSS strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.14 [2.28]	0.17 [2.71]	0.18 [2.81]	0.14 [2.18]	0.18 [2.79]	0.15 [2.29]
Quintile	NYSE	EW	0.23 [5.56]	0.25 [5.88]	0.24 [5.71]	0.23 [5.27]	0.24 [5.47]	0.23 [5.13]
Quintile	Name	VW	0.15 [2.21]	0.19 [2.77]	0.17 [2.58]	0.12 [1.75]	0.15 [2.28]	0.11 [1.63]
Quintile	Cap	VW	0.11 [1.74]	0.14 [2.15]	0.15 [2.23]	0.09 [1.35]	0.15 [2.18]	0.10 [1.47]
Decile	NYSE	VW	0.18 [2.20]	0.21 [2.54]	0.19 [2.28]	0.11 [1.34]	0.17 [2.00]	0.11 [1.26]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.09 [1.45]	0.12 [1.95]	0.13 [2.05]	0.11 [1.72]	0.13 [2.00]	0.11 [1.75]
Quintile	NYSE	EW	-0.02 [-0.42]					
Quintile	Name	VW	0.09 [1.38]	0.14 [2.00]	0.13 [1.86]	0.09 [1.40]	0.11 [1.60]	0.08 [1.26]
Quintile	Cap	VW	0.06 [0.99]	0.09 [1.35]	0.09 [1.44]	0.06 [0.94]	0.09 [1.40]	0.07 [1.01]
Decile	NYSE	VW	0.12 [1.43]	0.15 [1.81]	0.14 [1.60]	0.09 [1.08]	0.12 [1.38]	0.08 [0.99]

Table 3: Conditional sort on size and MSS

This table presents results for conditional double sorts on size and MSS. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on MSS. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high MSS and short stocks with low MSS .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	MSS Quintiles					MSS Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.53 [1.97]	0.83 [3.51]	0.90 [3.83]	0.88 [3.78]	0.72 [2.95]	0.19 [1.95]	0.20 [2.00]	0.18 [1.81]	0.21 [2.08]	0.17 [1.68]	0.20 [1.90]
	(2)	0.65 [2.79]	0.78 [3.54]	0.88 [4.07]	0.95 [4.28]	0.81 [3.59]	0.17 [2.31]	0.18 [2.51]	0.16 [2.16]	0.14 [1.88]	0.16 [2.18]	0.15 [1.94]
	(3)	0.66 [3.15]	0.77 [3.85]	0.79 [3.87]	0.89 [4.39]	0.74 [3.58]	0.08 [1.25]	0.09 [1.30]	0.07 [1.04]	0.07 [1.08]	0.07 [0.98]	0.07 [1.05]
	(4)	0.66 [3.41]	0.62 [3.31]	0.75 [3.86]	0.76 [3.95]	0.78 [4.16]	0.12 [1.92]	0.14 [2.24]	0.14 [2.15]	0.11 [1.79]	0.13 [2.01]	0.11 [1.72]
	(5)	0.51 [3.05]	0.52 [3.18]	0.55 [3.12]	0.62 [3.75]	0.58 [3.64]	0.07 [0.92]	0.10 [1.23]	0.10 [1.32]	0.05 [0.66]	0.11 [1.44]	0.07 [0.90]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	MSS Quintiles					MSS Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	374	379	380	380	377	28	35	34	35	28	
	(2)	108	108	108	107	108	56	56	56	56	56	
	(3)	79	79	79	78	78	99	98	97	99	98	
	(4)	67	67	67	67	67	215	208	205	209	216	
(5)	62	61	62	62	62	1695	1869	1306	1584	1695		

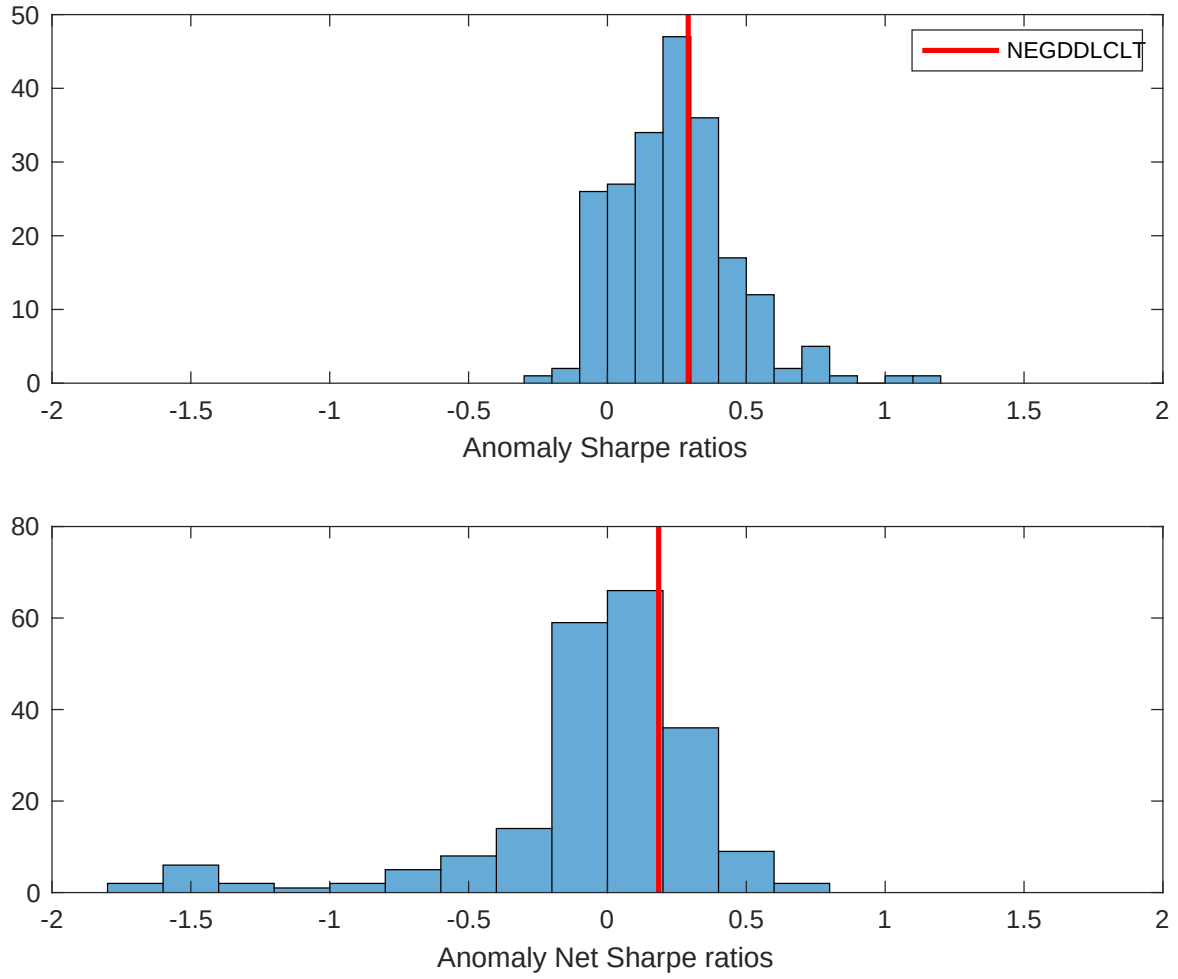


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the MSS with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

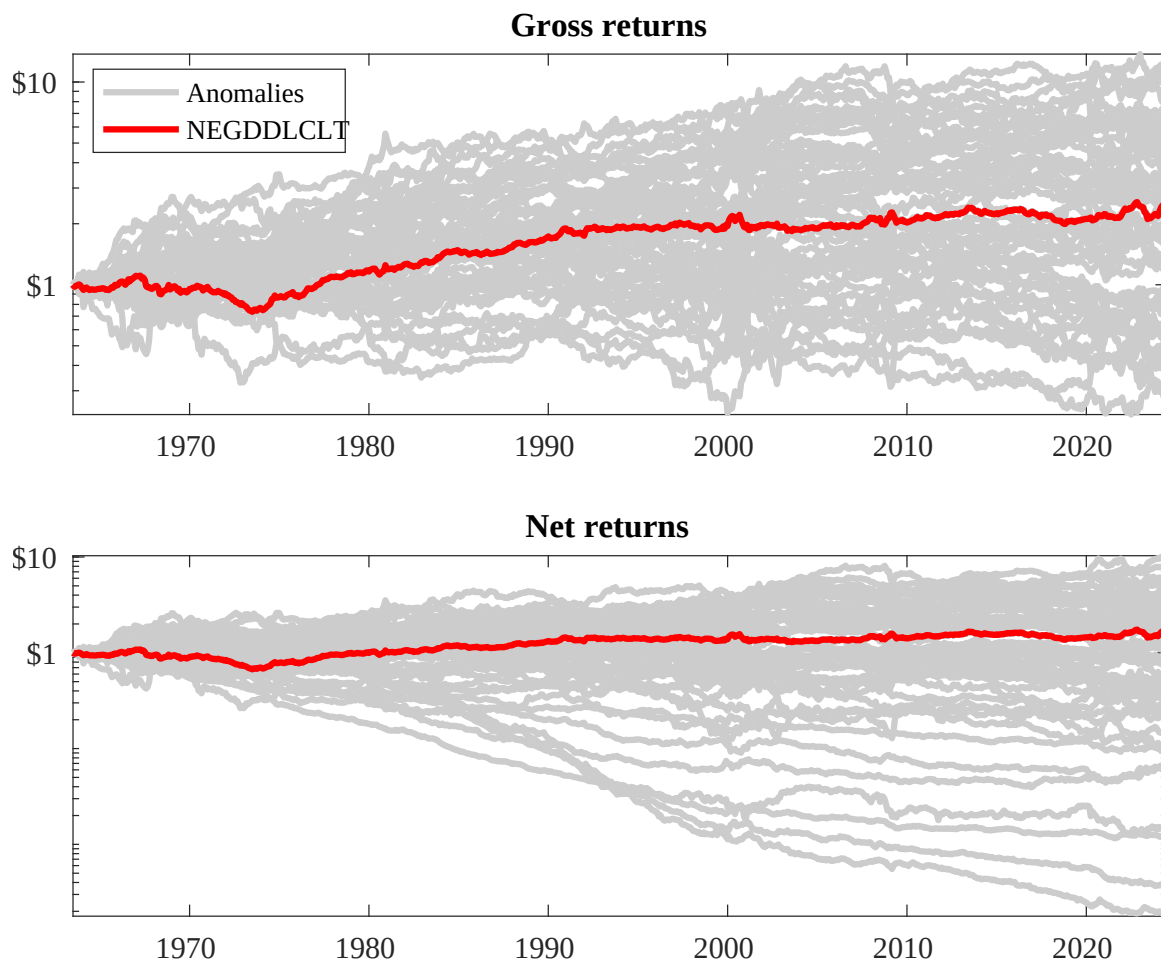
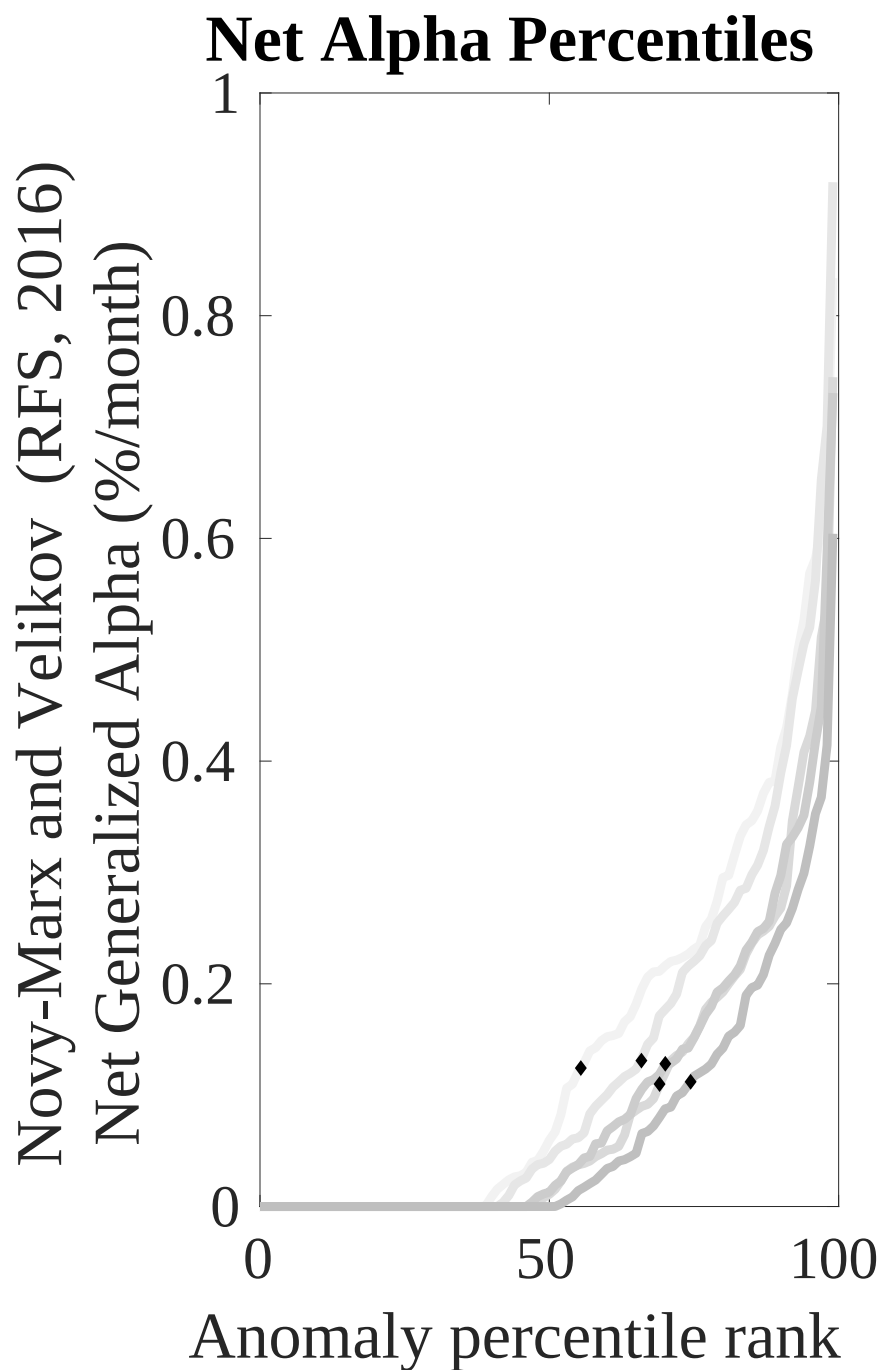
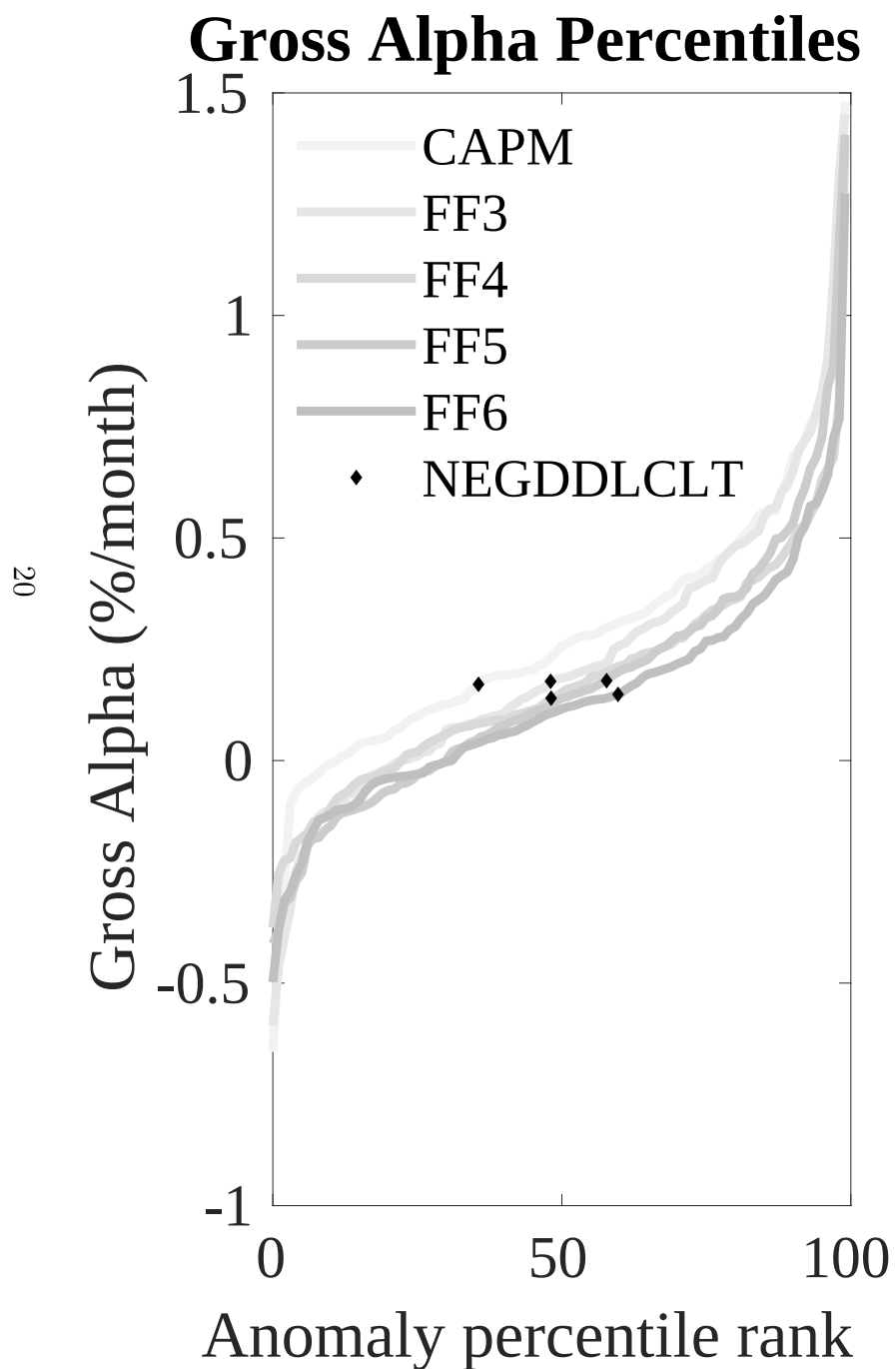


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the MSS trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



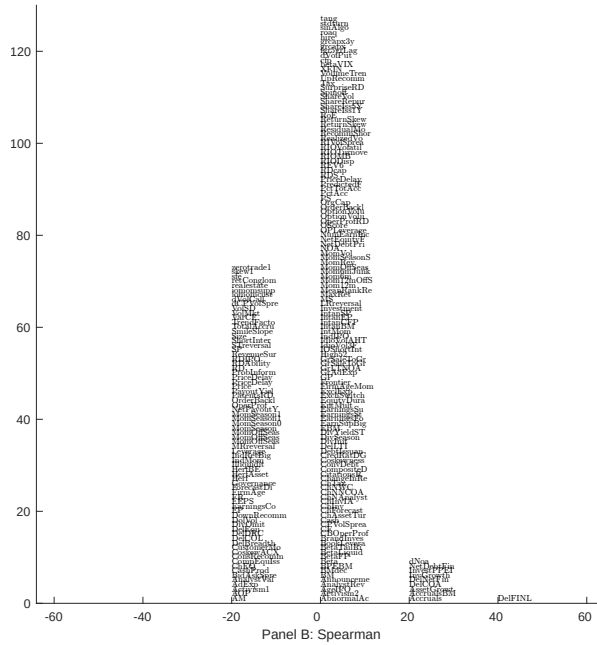
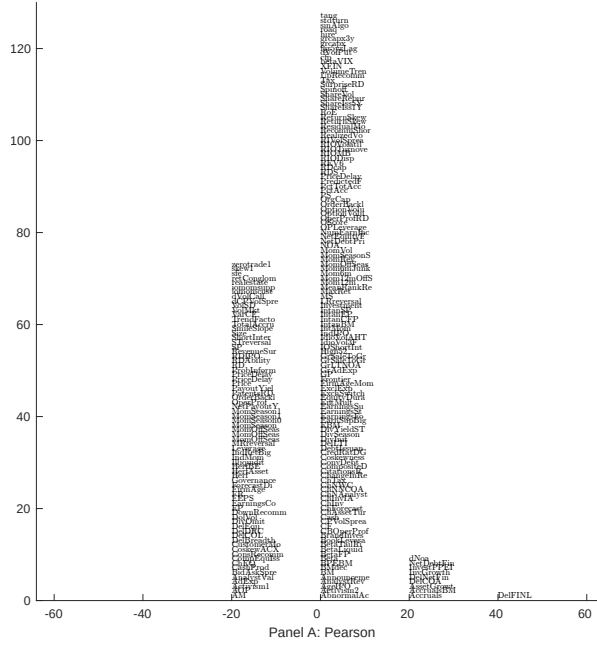


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with MSS. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

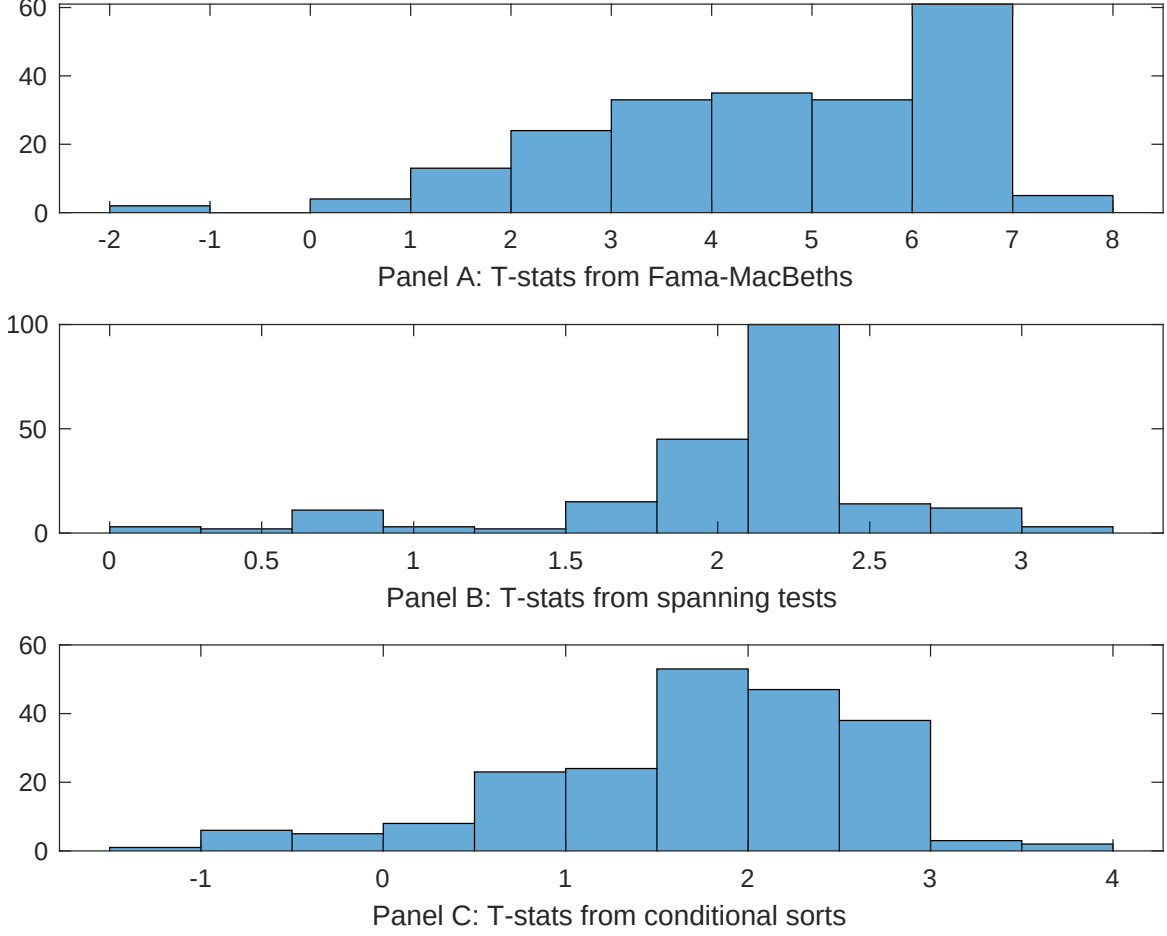


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of MSS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{MSS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{MSS}MSS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{MSS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on MSS. Stocks are finally grouped into five MSS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted MSS trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on MSS. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{MSS}MSS_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Accruals, Change in net financial assets, change in net operating assets, Change in Net Working Capital. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.12 [5.85]	0.13 [5.18]	0.12 [5.54]	0.12 [5.74]	0.13 [5.97]	0.12 [5.64]	0.13 [5.54]
MSS	0.46 [3.19]	0.75 [4.66]	0.74 [5.19]	0.69 [5.00]	0.36 [2.40]	0.86 [6.15]	0.53 [3.13]
Anomaly 1	0.16 [8.93]						0.76 [0.19]
Anomaly 2		0.19 [8.53]					0.11 [3.24]
Anomaly 3			0.13 [4.38]				0.57 [1.15]
Anomaly 4				0.74 [5.27]			-0.83 [-3.11]
Anomaly 5					0.13 [8.60]		0.14 [5.86]
Anomaly 6						0.60 [2.35]	-1.00 [-2.08]
# months	732	630	732	732	720	732	630
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the MSS trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{MSS} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Accruals, Change in net financial assets, change in net operating assets, Change in Net Working Capital. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.11 [1.90]	0.19 [2.86]	0.10 [1.55]	0.12 [1.89]	0.14 [2.12]	0.09 [1.50]	0.12 [1.99]
Anomaly 1	42.34 [12.27]						39.01 [8.03]
Anomaly 2		27.67 [7.71]					2.02 [0.48]
Anomaly 3			18.56 [7.24]				10.70 [4.21]
Anomaly 4				15.86 [5.12]			-6.46 [-2.01]
Anomaly 5					15.61 [4.31]		-3.48 [-0.94]
Anomaly 6						25.68 [8.22]	16.35 [4.93]
mkt	-2.89 [-2.03]	-4.08 [-2.69]	-2.47 [-1.63]	-3.65 [-2.39]	-3.85 [-2.50]	-4.18 [-2.80]	-2.80 [-2.03]
smb	-2.94 [-1.41]	0.63 [0.27]	5.19 [2.32]	2.36 [1.06]	1.61 [0.72]	2.42 [1.12]	1.03 [0.47]
hml	-4.03 [-1.46]	-5.61 [-1.93]	-3.57 [-1.21]	-7.41 [-2.50]	-8.63 [-2.89]	-5.46 [-1.89]	0.71 [0.26]
rmw	-12.53 [-4.50]	-9.87 [-3.31]	-2.61 [-0.85]	-6.85 [-2.27]	-8.85 [-2.95]	-6.24 [-2.13]	-6.17 [-2.16]
cma	-3.15 [-0.74]	1.39 [0.31]	7.58 [1.75]	15.77 [3.60]	1.35 [0.26]	11.73 [2.77]	-9.32 [-1.86]
umd	1.10 [0.77]	2.62 [1.72]	2.89 [1.92]	4.15 [2.73]	4.14 [2.71]	3.07 [2.06]	-0.27 [-0.19]
# months	732	630	732	732	732	732	630
$\bar{R}^2(\%)$	22	13	12	9	8	13	30

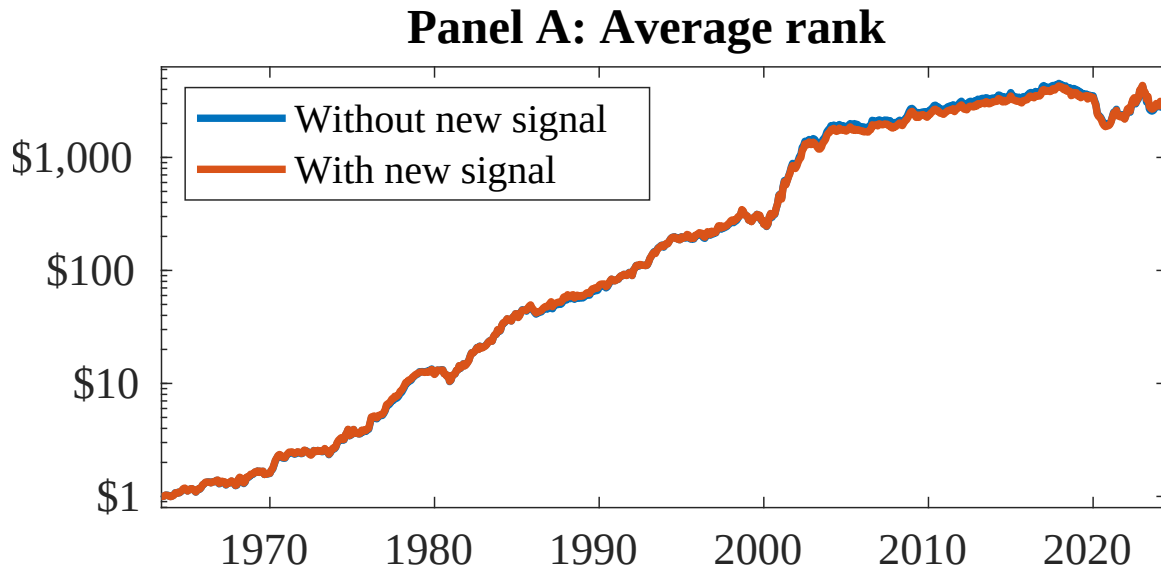


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as MSS. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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